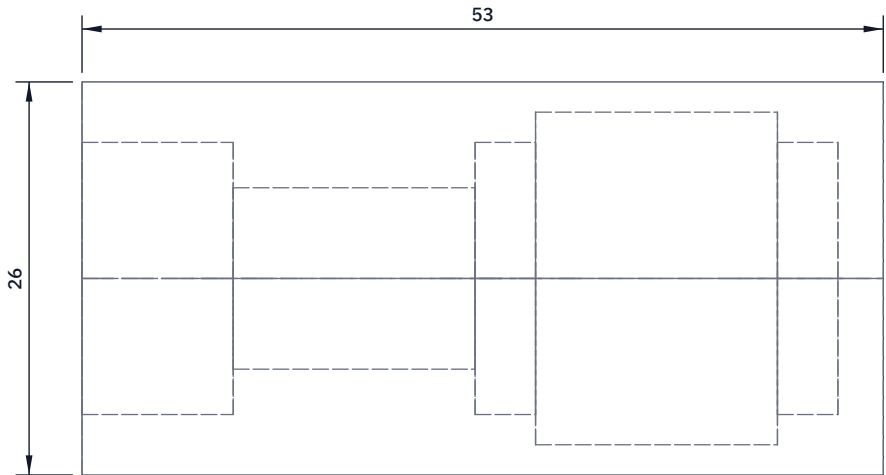
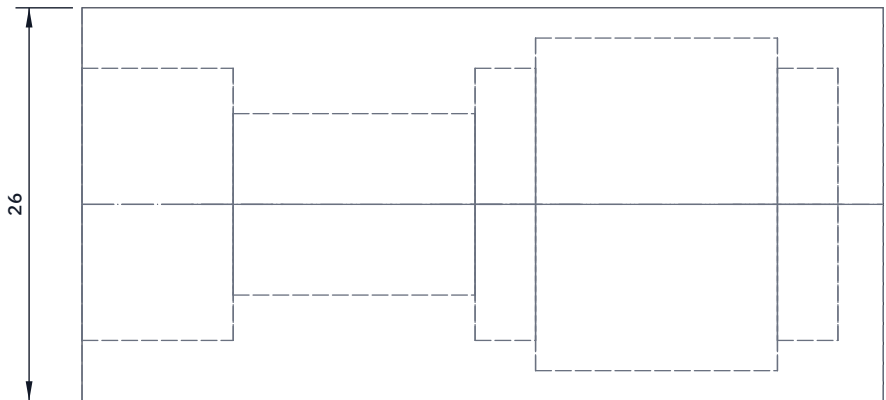


REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-09-30	AC



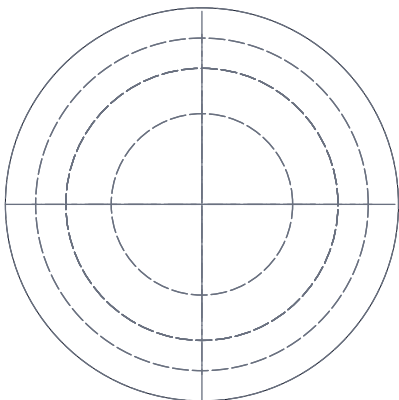
TOP VIEW

Scale 2:1; looking down (along -Z)



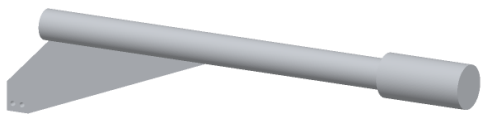
FRONT VIEW

Scale 2:1; looking at the front (along +Y)



RIGHT VIEW

Scale 2:1; looking at the right side (along -X)



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Diode housing: turn 26 mm bar to 30 mm long. Bore 18 mm, 10 deep, for the boom; bore 12 mm through for the diode module; counterbore 22 mm, 4 deep, at the front for the window spigot.

Window: 20 mm of 26 x 22 mm clear acrylic tube, plus a 4 mm spigot ring 22 x 18 mm solvent-welded inside each end.

End cap: turn 26 mm bar to a 3 mm disc with a 4 mm spigot 22 x 18; bond the cone mirror's stem to its centre, apex facing back.

The laser leaves the cone 300 mm in front of the dome centre: set this with the diode's focus and a card before bonding the cap.

Fit: boom bonded in the housing, diode clamped in its bore with a drop of silicone, window and cap bonded with clear epoxy.

Check: a sharp, even ring on a card held round the window.

CULVERTCRAWL

github.com/BoujeeEnjinia1701/culvertcrawl

CulvertCrawl laser head (housing, window, end cap): making sketch

DRAWING NO. CVC-DWG-111		REV P1	SHEET 1 of 1	SIZE ANSI B
SCALE 2:1	UNITS mm	PROJECTION 	DATE 2026-09-30	DRAWN Amish Chadha
MATERIAL / NOTES 6061 aluminium bar 26 mm; clear acrylic tube 26 x 22 mm				
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